Anuj Sharma

Department of Computer Science and Applications,
Panjab University, Chandigarh, India
E-mail: anujs AT pu.ac.in, anujsharma25 AT gmail.com
Website: https://www.anuj-sharma.in



Research Interests

Pattern Recognition and Machine Learning in the areas of Classification, Deep Learning, Document Analysis and Recognition, and Computational Optimization.

Education

Post-Doc from RWTH Aachen, Germany and PhD from Thapar Inst. of Engg. & Tech., India

Selected Publications

Journals

- K. Pathania, S. Singh and Anuj Sharma, "Transformer Models and their Cross-Domain Applications: A Systematic Review", Archives of Computational Methods in Engineering, Springer, 2026 (SCI, IF – 12.9)
- Anuj Sharma and P. Sharma, "FuzzyGAT: A Novel Node Classification Fuzzy Graph Attention Network with Uncertainty Handling", OPSEARCH, Springer, 2026 (SCI, IF – 1.7)
- S. Ratna, S. Singh and Anuj Sharma, "An Inclusive Analysis for Performance and Efficiency of Graph Neural Network Models for Node Classification", Computer Science Review, Vol. 56, 100722, Elsevier, 2025 (SCI, IF – 17.5)

- S. Singh, S Rohilla and Anuj Sharma, "An inclusive review on deep learning techniques and their scope in handwriting recognition", Vol. 24, No. 1, pp. 51-80, ELCVIA, 2025
- S. Kaur, Anuj Sharma and K. Dahiya, "Two-phase time minimization transportation problem with the restricted flow", Mathematics and Computers in Simulation, Vol. 229, pp. 611-635, Elsevier, 2024 (SCI, IF – 4.0)
- S. Kaur, E. Jain, Anuj Sharma, K. Dahiya, "An efficient algorithm for two-stage capacitated time minimization transportation problem with restricted flow", RAIRO - Operations Research, Vol. 58(4), pp. 2733-2766, Springer, 2024 (SCI, IF – 1.5)
- V. K. Chauhan, S. Singh and Anuj Sharma, “HCR-Net: A deep learning based script independent handwritten character recognition network”, Multimedia Tools and Applications, Vol. 83, pp. 78433–78467, Springer, 2024
- U. Thakur and Anuj Sharma, "Offline Handwritten Mathematical Recognition using Adversarial Learning and Transformers", International Journal of Document Analysis and Recognition (IJDAR), Vol. 27, pp. 147–158, Springer, 2023 (SCI, IF – 1.7)
- S. Singh, V. K. Chauhan and Anuj Sharma, "Indic Script Family and Its Offline Handwriting Recognition for Characters/ Digits and Words: A Comprehensive Survey", Artificial Intelligence Review, Vol. 56, pp. 3003–3055, Springer, 2023 (SCI, IF – 18.8)
- Anuj Sharma, S. Singh and S. Ratna, "Graph Neural Network Operators: a Review", Multimedia Tools and Applications, Vol. 83, pp 23413–23436, Springer, 2023
- E. Jain, K. Dahiya, Anuj Sharma, V. Verma, “Task preference-based bottleneck assignment problem”. Computational and Applied Mathematics, 41, 2022 (SCI IF – 2.3).

- F. Xie, Anuj Sharma and Z. Li, "An alternate approach to solve two-level priority based assignment problem", Computational Optimization and Applications, vol. 81, pp. 613-656, Springer, 2022 (SCI IF – 2.5)
- Charanjeet and Anuj Sharma, "A Review of Online Supervised Learning", Evolving Systems, Springer, 14, 343–364, 2022 (SCI, IF – 3.7)
- L. Kumari, S. Singh, V. V. S. Rathore, Anuj Sharma, 'Lexicon and attention based handwritten text recognition system", Machine Graphics and Vision, 31(1/4), 75–92, 2022.
- L. Kumari and Anuj Sharma, "A Review of Deep Learning Techniques in Document Image Word Spotting", Archives of Computational Methods in Engineering, vol. 29, pp. 1085-1106, Springer, 2021 (SCI, IF – 12.9)
- V. K. Chauhan, Anuj Sharma and K. Dahiya, " LIBS2ML: A library for scalable second order machine learning algorithms", Software Impacts, vol. 10, Elsevier, 2021 (SCI, IF – 1.5)
- S. Singh, Anuj Sharma and V. K. Chauhan, "Online handwritten Gurmukhi word recognition using fine-tuned Deep Convolutional Neural Network on offline features", Machine Learning with Applications, vol. 5, Elsevier, 2021 (SCI, IF – 6.1)
- J. Kaur, S. Singh and Anuj Sharma, "Dynamic features based stroke recognition system for signboard images of Gurmukhi text", Multimedia Tools and Applications, vol. 80, pp. 665–689 Springer, 2020
- V. K. Chauhan, Anuj Sharma and K. Dahiya, "Stochastic Trust Region Inexact Newton Method for Large-scale Machine Learning", International Journal of Machine Learning and Cybernetics, vol. 11, pp. 1541–1555, 2020 (SCI, IF – 2.9)
- Charanjeet and Anuj Sharma, "Modified online Newton step based on element wise multiplication", Computational Intelligence, vol. 36(3), pp. 1010-1025, 2020 (SCI, IF – 1.9)
- Charanjeet and Anuj Sharma, "Online learning using multiple times weight updating", Applied Artificial Intelligence, vol. 34(6), pp. 515-536, 2020 (SCI, IF – 4.0)

- P. Kaur, Anuj Sharma, V. Verma and K. Dahiya, "An alternate approach to solve two-level hierarchical time minimization transportation problem", 4OR, vol. 20, pp. 23-61, Springer, 2022 (SCI, IF – 2.0)
- S. Rohilla, Anuj Sharma and R. K. Singla, "Role of Sub-trajectories in Online Signature Verification", vol. 6, Array, Elsevier, 2020 (SCI, IF – 5.3)
- V. K. Chauhan, K. Dahiya and Anuj Sharma, "SAAGs: Biased Stochastic Variance Reduction Methods for Large-scale Learning", Applied Intelligence, vol. 49(9), pp 3331–3361, Springer, 2019 (SCI, IF – 3.5)
- S. Singh and Anuj Sharma, "Online Handwritten Gurmukhi Words Recognition: An Inclusive Study ", ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP), vol. 18(3), pp. 21:1-21:56, 2019, ISSN: 2375-4699 (SCI, IF – 2.7)
- E. Jain, K. Dahiya, Anuj Sharma and V. Verma, "An improved algorithm for two stage time minimization assignment problem", Journal of Combinatorial Optimization, vol. 37, pp. 713–736, 2019, ISSN: 1382-6905 (SCI, IF – 0.8)
- V. K. Chauhan, Anuj Sharma and K. Dahiya, "Faster Learning by Reduction of Data Access Time", Applied Intelligence, vol. 48, no. 12, pp. 4715-4729, 2019, ISSN: 0924-669X (SCI, IF – 3.5)
- V. K. Chauhan, K. Dahiya and Anuj Sharma, "Problem Formulations and Solvers in Linear SVM – a Review", Artificial Intelligence Review, vol. 52, pp. 803-855, 2019, ISSN: 0269-2821 (SCI, IF – 18.8)
- S. Singh, Anuj Sharma and I. Chhabra, "A Dominant Points based Feature Extraction Approach to Recognize Online Handwritten Strokes", International Journal of Document Analysis and Recognition (IJDAR), vol. 20, no. 1, pp. 37-58, 2017, ISSN: 1433-2825 (SCI, IF – 1.7)
- S. Rohilla, Anuj Sharma and R. K. Singla, "Optimal Feature Set Selection in Online Signature Verification", Int. J. Biometrics, vol. 9, no. 4, pp. 319-346, 2017, 1755-831X
- S Rohilla and Anuj Sharma, "SVM based Online Signature Verification Technique using Reference Feature Vector", Proceedings of the National

Academy of Sciences, India Section A: Physical Sciences, vol. 87, no. 1, pp. 125-136, 2017, ISSN: 0369-8203 (SCI, IF – 1.3)

- P. Kaur, Anuj Sharma, V. Verma and K. Dahiya, "A Priority based Assignment Problem", Applied Mathematical Modelling, vol. 40, no. 17-18, pp. 7784-7795, 2016, ISSN: 0307-904X (SCI, IF – 5.5)
- S Singh, Anuj Sharma and I. Chhabra, "Online Handwritten Gurmukhi Strokes Dataset based on Minimal Set of Words", Transactions on Asian and Low-Resource Language Information Processing (TALLIP), vol. 16, no. 1, 2016, ISSN: 2375-4699 (SCI, IF – 2.7)
- Anuj Sharma, P. Kaur, K. Dahiya and V. Verma, "An iterative algorithm for two level hierarchical time minimization transportation problem", European Journal of Operational Research, vol. 246, no. 3, pp. 700-707, 2015, ISSN: 0377-2217 (SCI, IF – 7.0)
- Anuj Sharma, "A Combined Static and Dynamic Feature Extraction Technique to recognize handwritten digits", Vietnam Journal of Computer Science, vol. 2 (3), pp. 133-142, 2015, ISSN: 2196-8888 (SCI, IF – 0.8)
- R. Kumar, Anuj Sharma and R. K. Sharma, "Recognition of Multi-Stroke Based Online Handwritten Gurmukhi Aksharas", Proceedings of the National Academy of Sciences, India Section A: Physical Sciences, vol. 85, no. 1, pp. 159-168, 2015, ISSN: 0369-8203, (SCI, IF – 1.3)
- Anuj Sharma, R. Kumar and R. K. Sharma, "HMM Based Online Handwritten Gurmukhi Character Recognition", International Journal of Machine Graphics and Vision, vol. 19, no. 4, pp. 439-449, 2010, ISSN 1230-0535
- Anuj Sharma, R. Kumar and R. K. Sharma, "Online preprocessing of handwritten Gurmukhi strokes", International Journal of Machine Graphics and Vision, vol. 18, no. 1, pp. 115-120, 2009, ISSN 1230-0535

Conference Proceedings/ Chapters

- U. Thakur and Anuj Sharma, "Newton Methods-Based Convolution Neural Networks Using Parallel Processing", Proceedings of the NIELIT's

International Conference on Communication, Electronics and Digital Technologies, Lecture Notes in Networks and Systems, vol. 1, pp. 121-136, Springer, 2025

- L. Kumari, S. Singh, VVS Rathore and Anuj Sharma, "A Comprehensive Handwritten Paragraph Text Recognition System: LexiconNet", Workshop on Machine Learning, International Conference on Document Analysis and Recognition (ICDAR), Lecture Notes in Computer Science (LNCS), Springer, 2023
- L. Kumari, S. Singh, VVS Rathore and Anuj Sharma, "A Lexicon and Depth-wise Separable Convolution Based Handwritten Text Recognition System ", Proceedings of IVCNZ 2022 (an IAPR conference), Lecture Notes in Computer Science (LNCS), Springer, 2022
- L. Kumari, S. Singh and Anuj Sharma, "Page Level Input for Handwritten Text Recognition in Document Images", In proceedings of 7th International Conference on Harmony search, Soft computing and Applications (ICHSA), Seoul, South Korea 2022
- Charanjeet and Anuj Sharma, "Online Newton step based on pseudo inverse and elementwise multiplication", In proceedings of 7th International Conference on Harmony search, Soft computing and Applications (ICHSA), Seoul, South Korea, 2022
- J. Kaur, S. Singh and Anuj Sharma, "Recognition of Gurmukhi signboard image classes based on static and dynamic feature sets", proceedings of 7th National Conference on Computer Vision, Pattern Recognition, Image Processing and Graphics (NCVPRIPG 2019), Communications in Computer and Information Science, 2019 (Scopus indexed book series by Springer publisher)
- S. Rohilla, Anuj Sharma and R. K. Singla, "Mobile Based Signature Database (MOBSIGN DB): A new signature database", proceedings of International Conference on Advanced Communication and Computational Technology (ICACCT-2019), Advances in Communication

and Computational Technology, 2019 (Scopus indexed book series by Springer publisher)

- S. Singh and Anuj Sharma, "Recognition of online handwritten Gurmukhi characters through neural networks", proceedings of International Conference on Advanced Communication and Computational Technology (ICACCT-2019), Advances in Communication and Computational Technology, 2019 (Scopus indexed book series by Springer publisher)
- Charanjeet, S. Singh and Anuj Sharma, "A comparative analysis on Online Handwritten Stroke Classification using online learning", proceedings of International Conference on Advanced Communication & Computational Technology (ICACCT-2019), Advances in Communication and Computational Technology, 2019 (Scopus indexed book series by Springer publisher)
- A. Kumar, Anuj Sharma and R. K. Singla, "An Efficient License Plate Extraction Technique", Proceedings of International Conference on Recent Advancement in Computer, Communication and Computational Sciences (RACCCS-2018), Ambient Communications and Computer Systems, (a chapter), pp. 487-495, 2018 (Scopus indexed book series by Springer publisher)
- V. K. Chauhan, K. Dahiya and Anuj Sharma, "Mini-batch Block-coordinate based Stochastic Average Adjusted Gradient Methods to Solve Big Data Problems", Proceedings of Machine Learning Research (PMLR), vol. 77, pp. 49-64, 2017, ISSN: 2640-3498 (A reputed ML proceedings publisher)
- V. K. Chauhan, K. Dahiya and Anuj Sharma, "Trust Region Levenberg-Marquardt Method for Linear SVM", IEEE Proceedings of the Ninth International Conference on Advances in Pattern Recognition (ICAPR-2017), Indian Statistical Institute, Bangalore
- K. Dahiya, Anuj Sharma and V. Kumar, “Online Support Vector Machine Based on Minimum Euclidean Distance”, proceedings of International Conference on Computer Vision and Image Processing (CVIP), Advances in

Intelligent Systems and Computing (AISC), ISSN: 2194-5357, 2016 (IAPR Conference and in scopus indexed book series by Springer publisher)

- S. Rohilla, Anuj Sharma and R. K. Singla, "Online Signature Verification at Sub-trajectory Level", proceedings of 2nd International Conference on Advanced Computing, Networking and Informatics (ICACNI-2014), Advanced Computing, Networking and Informatics in Series of Smart Innovation, Systems and Technologies, vol. 28, pp. 369-374, 2014, ISSN: 2190-3018 (published in scopus indexed book series by Springer publisher)
- K, Dahiya and Anuj Sharma, “Reducing Neural Networks training data using Support Vectors” In IEEE Proceedings of Recent Advances in Engineering and Computational Sciences (RAECS) 2014, pp. 1-4, ISBN: 978-1-4799-2290-1
- Anuj Sharma, “Recovery of Drawing order in Handwritten Digit Images”, In IEEE Proceedings of Second International Conference on Image Information Processing (ICIIP) 2013, pp 437-441, ISBN: 978-1-4673-6099-9
- Anuj Sharma, R. Kumar and R. K. Sharma, “Rearrangement of Strokes in Recognition of Online Handwritten Gurmukhi Words", IEEE Proceedings of 10th International Conference on Document Analysis and Recognition, Barcelona, Spain (ICDAR-2009), pp. 1241-1245, ISBN: 978-0-7695-3725-2 (IAPR Conference)
- Anuj Sharma, R. Kumar and R. K. Sharma, "Recognizing Online Handwritten Gurmukhi Characters using Elastic Matching", IEEE Proceedings of International Congress on Image and Signal Processing (CISP-2008), Sanya, vol. 2, pp. 391-396, ISBN: 978-0-7695-3119-9

Text Book

- Anuj Sharma, "Essentials of AI and Soft Computing", PHI Learning, 2024 (Book)

Invited Talks

- "From Standard Classifiers to Attention-Driven Models" in Short Term Course "Research in the Age of Artificial Intelligence: Methods, Ethics and Impact" April 2026
- Invited Talk in 29th Punjab Science Congress "Cutting Edge Science & Technology: Empowering Youth for Viksit Bharat", Feb 2026
- Talk in AICTE Faculty Development Programme, MCMDAV College Chandigarh, Dec 2025
- "Decoding Data: The Power of Classification Algorithms in ML" in Skill Enhancement Course in Machine Learning using Python Oct 2025
- "Introduction to Cyber Safety and Security" in Cyber Awareness Workshop at MGSIPA, Chandigarh, July-2025
- "Introduction to Artificial Intelligence and Machine Learning" and "Use of AI in Education/ Teaching", in Faculty Development Programme at GC College Sanjauli, Shimla, May 2025
- "Data Analytics in Higher Education", in Short Term Course on AI in Higher Education organized by UGC-MMTTC, Panjab University, Chandigarh, Feb 2025
- Delivered various talks as CBSE Resource person on "Use of AI in Education", 2025
- "Introduction to AI and Machine Learning" in Workshop on Artifiicial Intelligence and Machine Learning, UIAMS, Panjab University Chandigarh, April 2024
- Delivered various talks as CBSE Resource person on "Cyber Security",s "Python Language" and “Role of Artificial Intelligence in Teaching”, 2024
- "Introduction to Artificial Intelligence and Machine Learning " and "Data Prediction using Machine Learning Techniques", special lecture, Department of Library and Information Science, Panjab University, Chandigarh, February 2024
- "Machine Learning and its Applications" and "Artificial Intelligence and its Real Life Applications", in Skill Development Course on ‘Artificial

Intelligence and Machine Learning' funded by RUSA, Department of Computer Science and Applications, Panjab University, Chandigarh, August 2023

- "Classes In Python Programming", in Skill Development Course funded by RUSA, Department of Computer Science and Applications, Panjab University, Chandigarh, May 2023
- "Introduction to Classification" and "Introduction to Deep Learning", in Skill Development Course funded by RUSA, Department of Computer Science and Applications, Panjab University, Chandigarh, April 2023
- "Introduction to Support Vector based Classification" and "SV based classification in Batch based and Online Learning System", Department of Data Science and Artificial Intelligence, Himachal Pradesh University, Shimla, 9-Dec-2022
- "Python Libraries and Plotting", 7-Days National Workshop (Online)/Value Added Course on Open Source Software Utility to Applied Econometrics and Statistics, conducted by the Department of Economics in collaboration with the Department of Statistics & the Department of Computer Science and Applications, Panjab University, Chandigarh. Nov 2022
- "Data Science (AI/ML) and its use in Real Life Applications", Narsee Monjee Institute of Management Studies (NMIMS), Deemed University, Chandigarh, August 2022
- "Scope of Artificial Intelligence", Dainik Bhaskar Newspaper - Education Career Fair, July 2022, Chandigarh (One session of 60 minutes)
- "Role of AI and ML in real life applications", Short term attachment programme, Center for Innovation ad Bio-Design (CIBioD), PGIMER, Chandigarh, June 2022
- "ML based Regression and Classification Implementation", One Day workshop on Machine Learning Techniques, Department of Computer Science and Applications, Panjab University, Chandigarh 4-Feb-2022

- "Avenues of Application of Artificial Intelligence in Health and Medtech", Short term attachment programme, Center for Innovation ad Bio-Design (CIBioD), PGIMER, Chandigarh, May 2021
- "Support Vector Machine Classifiers", Refresher course in Information communication and Technology, HRDC, Panjab University, Chandigarh, Feb 2021
- "Role of AI in Education and Machine Learning Techniques to Solve Teacher Perspective Tasks", Workshop in Application of Artificial Intelligence in Teaching and Learning, Institute for Development and Comunication, Chandigarh, Feb 2021
- “Artificial Intelligence and Machine Learning”, Faculty Development Programme in “Artificial Intelligence and Machine Learning – Basics and Applications”, CALEM, Panjab University, Chandigarh, 22-28 Dec, 2020
- “Introduction to Artificial Intelligence and Machine Learning”, Faculty Development Programme in Artificial Intelligence in Human Life, HRDC, Panjab University, Chandigarh, 6-12 Oct, 2020
- “Advanced Python and Machine Learning”, Scientific Computing using Matlab and Python, JCBUST, YMCA, Faridabad, 14-18 Oct 2020
- "Working with Python and Python use in Machine Learning", Department of Statistics, Panjab University, Chandigarh, 11 March, 2020
- “Fundamentals of Programming in Matlab”, Workshop on Mathematical Programming, MCM DAV College, Chandigarh, 1-2 Nov 2018
- “Graphics Programming in Matlab”, International Workshop on Soft Mathematical Tools, Khalsa College Amritsar Talk, 17-April-2018
- “Solving Large Scale Machine Learning Problems”, GNEC Ludhiana, 7-March- 2018
- "Fuzzy Sets and Membership Functions", National Institute of Technical Teachers Training and Research Institute, Chandigarh, 20 Sep 2017
- “Programming in Matlab”, SASE, DRDO, Chandigarh, 9-Sep-2016
- “Classifiers as Support Vector Machines and Neural Networks”, National Conference at DSCW, Ferozpur, March 2016.

- "Fuzzy Programming", National Institute of Technical Teachers Training and Research Institute, Chandigarh, 24 Feb 2016
- "Coding Theory", National Institute of Technical Teachers Training and Research Institute, Chandigarh, 25 Feb 2016
- "Computer programming for Mathematics Problems using Matlab", Instructional School of Lecturers in Numerical Analysis (Sponsored by National Centre of Mathematics), Panjab University, Chandigarh, (9-28) June 2014.
- "Matlab tutorials", National Institute of Technical Teachers Training and Research Institute, Chandigarh, (27-31) May 2013.
- "A Feed Forward Neural Network Recognition System using Efficient Features Selection", Mathematics Department, Panjab University, Chandigarh Science Congress, Feb, 2012.
- "Neural Networks Applications in Computer Vision", Mathematics Department, Panjab University, Symposium Feb 7-8, 2012.

Misc.

- Acted as Convener/ Coordinator/ Participant/ Attendee in many academic conferences/ seminars/ programmes.

(*for updates, please visit home page at*: https://www.anuj-sharma.in)